

SMART LIBRARY REQUEST WORKFLOW IN SERVICENOW

Project Title: Smart Library Request Workflow in ServiceNow

Team lead :Koilapu.Sravani

Team members: Mandala.Sireesha

Nasa.Tulasi

Tungana.Niharika

Yenduva.Devi

Course: ServiceNow System Administrator

Platform: ServiceNow Personal Developer Instance (PDI)

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1. Introduction

The Smart Library Request Workflow is a ServiceNow application developed to automate the library book borrowing and return process. Instead of maintaining manual records, the system allows students to request books online while librarians manage approvals, book issuance, and returns. The workflow improves efficiency, reduces paperwork, and ensures accurate record management.

Smart Library Request Workflow in ServiceNow

Team

Overview

Workspace

Kanban

Requirement Analysis & Planning

Epic

Complexity: Hard

Duration: 2h 0m

Description

The system requires two roles: student and librarian. Students can request and track books, while librarians manage the catalog and approve requests. Two tables are needed: **Book** (details, status) and Borrow Request (transactions). Workflows will automate approvals and status updates, ACLs will control access, and reports like "Most Borrowed Books" will give insights. Planning covers defining tables, roles, workflows, and test cases to ensure smooth implementation.

2. Project Objectives

The objectives of this project are:

Automate the library book borrowing process.

Reduce manual work for librarians.

Maintain accurate book availability records.

Implement role-based access for users.

Automate approvals using Flow Designer.

Generate reports for library management.

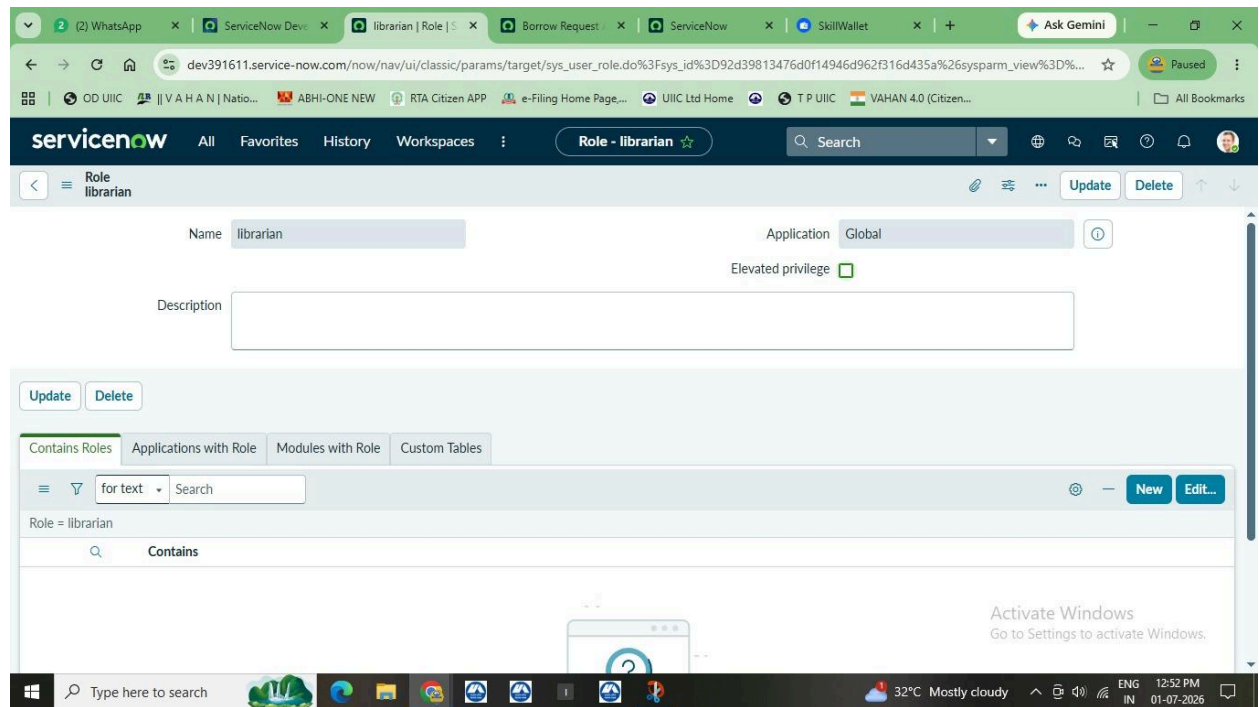
3. Project Description

The application consists of two users: Student and Librarian. Students submit book requests, and librarians review and approve them. Once approved, the book is issued, and after it is returned, the book becomes available again. The system records every transaction and allows reports to be generated for monitoring library usage.

4. Roles Creation

- Two roles were created in the application.
- Librarian Role
- The Librarian role has permission to:
 - Add new books.
 - Update book details.
 - Approve or reject requests.
 - Issue books.
 - Receive returned books.
 - View reports.
- Student Role
- The Student role has permission to:
 - Request books.
 - View request status.
 - Return borrowed books.
 - Navigate to User Administration.
 - Open Roles.
 - Click New.
 - Create Librarian role.
 - Create Student role.
 - Save the roles.

Insert Screenshot: Roles Creation



The screenshot shows the ServiceNow interface for creating or editing a role. The browser address bar indicates the URL: `dev391611.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3D92d39813476d0f14946d962f316d435a%26sysparm_view%3D%26sysparm_target%3Droles`. The page title is "Role - librarian". The form includes fields for "Name" (librarian), "Application" (Global), and "Elevated privilege" (unchecked). There is a "Description" text area. Below the form are "Update" and "Delete" buttons. A tabbed interface shows "Contains Roles" as the active tab, with a search bar and "New" and "Edit..." buttons. The "Contains" table is empty. The Windows taskbar at the bottom shows the date and time as 12:52 PM on 01-07-2026.

5. Table Creation

Two custom tables were created.

Book Table

The Book table stores information about books available in the library.

Fields include:

Book Name

Author

Category

Availability

Total Copies

Borrow Request Table

The Borrow Request table stores information about student book requests.

Fields include:

Request Number

Student Name

Book

Request Date

Issue Date

Return Date

Status

Steps Performed

Navigate to System Definition.

Open Tables.

Create Book table.

Create Borrow Request table.

Add required fields.
Save both tables.

The screenshot displays the ServiceNow interface for configuring two tables. The top section shows the 'Table - Borrow Request' configuration page. The 'Columns' tab is active, showing a list of dictionary entries for the 'Borrow Request' table. The bottom section shows the 'Table - book' configuration page. The 'Columns' tab is active, showing a list of dictionary entries for the 'book' table. The 'book' table configuration includes fields for 'Label' (set to 'book'), 'Name' (set to 'u_book'), 'Application' (set to 'Global'), and 'Remote Table' (set to 'false').

Table - Borrow Request

Column label	Type	Reference	Max length	Default value	Display
Updated by	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updates	Integer	(empty)	40		false
Request Data	Date/Time	Created time	40		false
Updated	Date/Time	(empty)	40		false
Status	Choice	Requested Item	40		false
Return Date	Date/Time	(empty)	40		false
Created by	String	(empty)	40		false

Table - book

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label: Application:

* Name: Remote Table: ☐

Table - book

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Created by	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Title	String	(empty)	40		false

6. Field Configuration

Appropriate fields were configured for each table.

Book Table includes:

Book Name

Author

Category

Availability

Borrow Request Table includes:

Student

Book

Request Date

Issue Date

Return Date

Status

The correct field types such as String, Reference, Date, Boolean, and Choice were used

The screenshot shows the ServiceNow UI Policy configuration interface. The browser address bar displays the URL: `dev391611.service-now.com/now/nav/ui/classic/params/target/sys_ui_policy.do%3Fsys_id%3D1dc6996747e1cf14946d962f316d436c%26sysparm_view%3D%2...`. The page title is "UI Policy - Make Return Date manda...". The configuration details are as follows:

- Table:** Borrow Request [u_borrow_reque...]
- Application:** Global
- Active:** ☒
- * Short description:** Make Return Date mandatory when issued
- When to Apply:**
 - Conditions:** Add Filter Condition, Add OR Clause
 - Condition 1: Status is Approved
 - Logic: AND

Buttons for "Update" and "Delete" are visible at the bottom of the configuration section. An "Activate Windows" watermark is present in the bottom right corner of the screenshot.

7. Reference Qualifier

A Reference Qualifier was configured for the Book reference field.

Purpose:

Only books that are available are displayed to students while creating a borrow request.

This prevents unavailable books from being selected.

8. Flow Designer Configuration

Flow Designer was used to automate the approval process.

Workflow:

Student submits request.

Librarian receives approval task.

Librarian approves request.

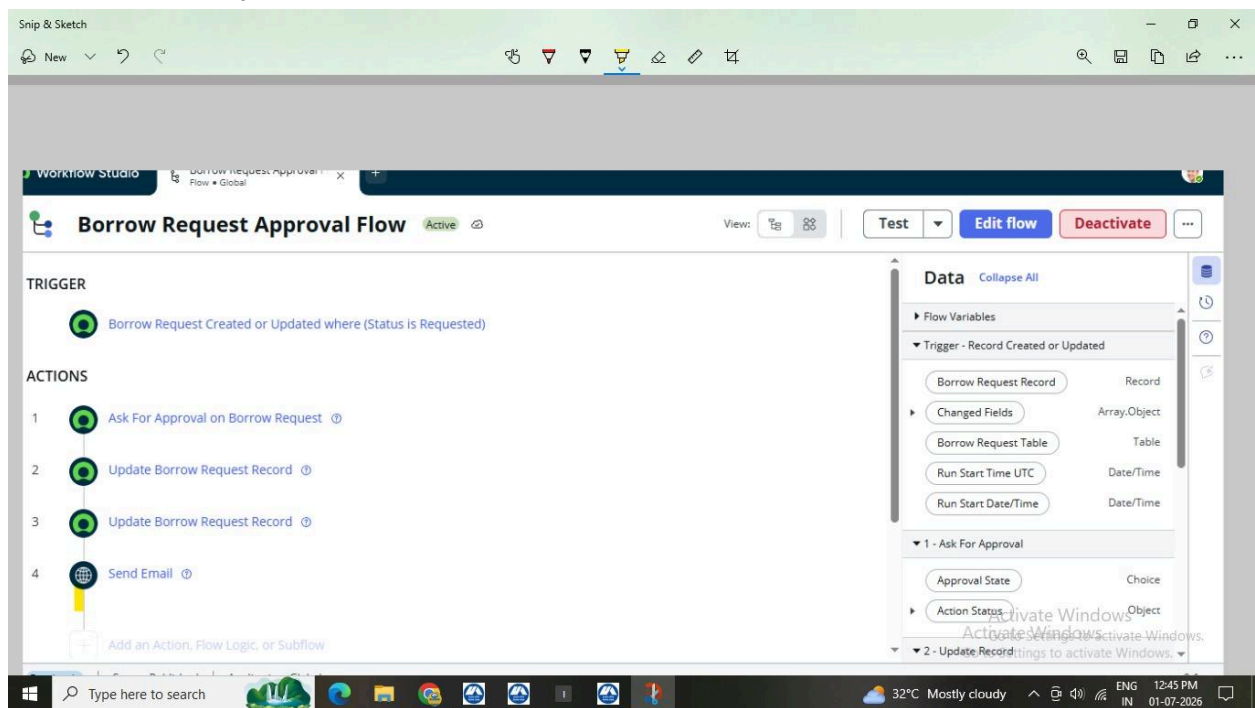
Book is issued.

Student returns book.

Book availability is updated automatically.

Benefits:

Eliminates manual approval.
Reduces processing time.
Improves efficiency.



9. UI Policy Configuration

A UI Policy was created.

Purpose:

When the book status becomes Returned, the Return Date field becomes mandatory.
This ensures that every returned book has a valid return date recorded.

Insert Screenshot: UI Policy

The screenshot shows the ServiceNow UI Policy configuration interface. The browser address bar indicates the URL: `dev391611.servicenow.com/now/nav/ui/classic/params/target/sys_ui_policy.do%3Fsys_id%3D1dc6996747e1cf14946d962f316d436c%26sysparm_view%3D%2...`. The page title is "UI Policy - Make Return Date manda...".

UI Policy Configuration Details:

- Table:** Borrow Request [u_borrow_reque...]
- Application:** Global
- Active:** ☒
- * Short description:** Make Return Date mandatory when issued

When to Apply

Conditions: [Add Filter Condition](#) [Add OR Clause](#)

Status is Approved [AND](#) [OR](#) [X](#)

[Update](#) [Delete](#)

Activate Windows
Go to Settings to activate Windows.

10. Access Control List (ACL)

ACLs were implemented to secure the application.

Students can:

Create requests.

View their own requests.

Students cannot:

Approve requests.

Delete records.

Librarians can:

Manage all requests.

Update records.

Issue books.

Return books.

Generate reports.

The screenshot shows the 'Access Control - New Record' form in ServiceNow. The form includes fields for Type (record), Operation (read), Application (Global), Active (checked), Admin overrides (checked), Decision Type (Allow If), Protection policy (-- None --), Name (book [u_book]), and Description (Default Access control on u_book). A warning message at the top states: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' The bottom of the form shows the 'Conditions' section with a message: 'Access Control Rules have two decision types, and these types will behave differently depending on conditions. 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access. 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may grant access to records where this ACL has denied access.'

11. Report Creation

A report named "Most Borrowed Books" was created.

Purpose:

Identify popular books.

Monitor borrowing trends.

Help librarians manage inventory.

The screenshot shows the 'Report - Most Borrowed Books' configuration page in ServiceNow. The form includes fields for Application (Global), Title (Most Borrowed Books), Table (Borrow Request [u_borrow_request]), Field Name (u_book), Type (Vertical bar), Chart size (Large), User (6816f79cc0a8016401c5a33be04be441), Filter (Add Filter Condition, Add OR Clause), and Roles (edit icon). The bottom of the form shows the 'Update', 'View Report', 'Edit Report', and 'Delete' buttons. A message at the top states: 'Please configure this report using Edit Report, and create new reports using View / Run.'

12. Testing

The application was tested successfully.

Test Case 1

Student submits request.

Result: Request created successfully.

Test Case 2

Librarian approves request.

Result: Status changed to Approved.

Test Case 3

Book issued.

Result: Book availability changed to False.

Test Case 4

Book returned.

Result: Book availability updated to True.

13. Technical Blueprint

Application Workflow

Student



Submit Book Request



Borrow Request Table



Approval by Librarian



Issue Book



Return Book



Book Availability Updated



Reports Generated

14. Setup Manual

Follow these steps to recreate the project:

Login to ServiceNow Personal Developer Instance.

- Create the Librarian role.
- Create the Student role.
- Create the Book table.
- Create the Borrow Request table.
- Configure all required fields.
- Configure Reference Qualifier.
- Create UI Policy.
- Configure Access Control Lists.
- Create Flow Designer workflow.
- Create reports.
- Test the complete workflow.

15. Conclusion

.In conclusion, anchoring a smart library workflow within ServiceNow does more than just digitize the service desk—it creates a highly visible, measurable ecosystem. Librarians transition from administrative gatekeepers to data-driven service managers who can track resource lifecycle trends, predict bottlenecks, and optimize budgets. Ultimately, ServiceNow ensures the library remains a highly efficient, sustainable, and reliable asset for the entire community.”